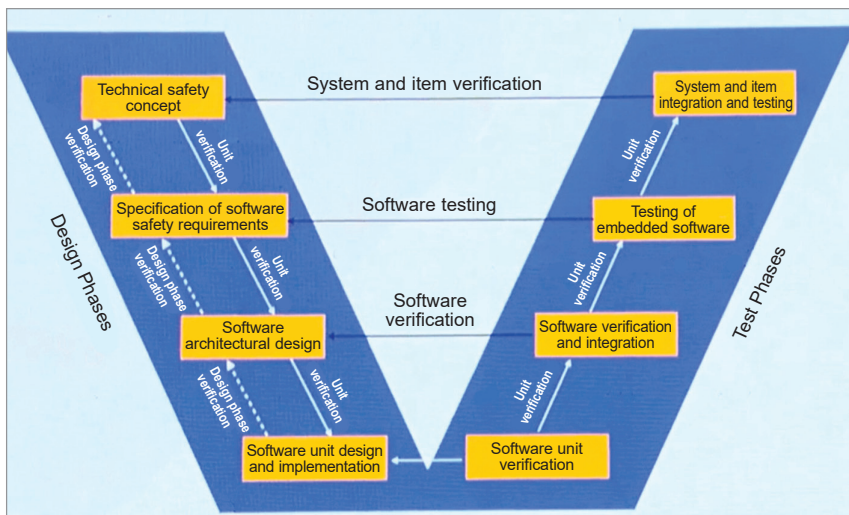




Safety standards, ISO 26262

in this article is on two-wheelers. Currently, most two-wheeler manufacturers prioritise basic control systems. However, many companies are working on advanced systems that promise more sophisticated features, including collision detection, emergency braking, distance warnings, and balance maintenance.

Despite the temptation to consolidate all intelligence into a single motor control unit (MCU) or battery management system (BMS), this approach has its challenges. MCUs and BMSs are already responsible for critical functions within a vehicle, such as controlling the powertrain and managing the battery. Centralising all subsystems into these units can lead to complications and significantly increase the risk of severe failures in the event of system malfunctions.

The role of vehicle control units

To succeed, manufacturers must prioritise safety, adhere to global standards, and adopt innovative solutions like vehicle control units (VCUs) and AUTOSAR (automotive open system architecture). By doing so, India can lead the way in the supply chain, delivering cutting-edge features, reliability, and safety

to discerning consumers worldwide.

VCUs have emerged as a central communication hub to address this challenge. They coordinate and manage critical components like motor controllers and battery management systems while facilitating communication with other subsystems. This approach enhances overall vehicle safety and intelligence.

In India, it is imperative to prioritise safety and quality for the automotive industry. Establishing high standards in the design, production, and verification of electronic systems is vital. Adhering to international safety standards, such as ISO 26262, is essential for ensuring the reliability and safety of automotive control systems.

ASIL, or Automotive Software Integrity Level, categorises the reliability and safety of vehicle systems. It considers factors like failures' criticality, frequency, and ease of control. Systems are classified into four categories: A, B, C, and D, with A being the most critical. ASIL guides the development process, helping manufacturers design systems that meet safety and reliability requirements.

In 2025, 2-wheelers are expected to be equipped with advanced features and electronic systems to

enhance rider safety and convenience. These include an advanced rider assistance system (ARAS) encompassing features like a traction control system, remote passive keyless ignition, body control module, lighting controller, and semi-active suspension. Additionally, on-board diagnostics, navigation, and telematics will be integrated into the 2-wheelers, along with GPS-USB charging for mobile devices and smartphone connectivity. Other notable features comprise an electronic steering damper, bank angle display, and infotainment functionalities. In total, these vehicles will boast 20-30 electronics features, accounting for approximately 20-25% of the overall vehicle cost.

AUTOSAR: A revolution in EV development

To streamline the development process and ensure effective communication between subsystems, the industry relies on AUTOSAR. AUTOSAR offers a standardised approach to mapping functions and logic, simplifying the integration of diverse components and systems. This advanced software platform has become the backbone of modern EVs, enabling a level of flexibility, standardisation, and reliability.

Traditionally, each component had its proprietary protocols and interfaces when developing automotive systems. This lack of standardisation made it challenging for different systems to communicate effectively, leading to inefficiencies and complexities in the vehicle's architecture.

AUTOSAR changes this paradigm by providing a common framework for developing automotive software. It integrates various functions into a single software stack, simplifying the development process and ensuring compatibility between different components and systems. This approach allows automakers to